

Artificial Intelligence

Semester Project Brief

WORTH 25% OF MODULE
GRADE



01 Overview

This is an individual project where you collect data from a website, explore it, and try out a few machine learning techniques on it. You will put everything in one Jupyter notebook and write a short report about what you found. Everything is due in Week 13.

02 Data

Use Python to scrape data from a public website. You need at least 200 rows and 6 columns. Ready-made datasets from places like Kaggle are not allowed. Save your data as a CSV file and write a short note about where it came from and what the columns mean.

03 Tasks

Do the following in a single Jupyter notebook. Use clear headings to separate each section.

3.1 Data Preparation

Clean the data by handling any missing values and removing duplicates. Convert text columns to numbers where needed, and split your data into a training set (80%) and a testing set (20%). Show some basic statistics and include at least two simple plots that help you understand the data.

3.2 Regression

Pick a numeric column you want to predict, such as a price or a rating, and train a linear regression model on it. Show how well it did by reporting the error and plotting the predicted values next to the actual ones.

3.3 Classification

Take the same column and split it into two or three groups (for example, low and high). Then train a classifier to predict which group each item belongs to, and show a confusion matrix to see where it got things right and where it made mistakes.

3.4 Clustering

Use k-means to group your data into clusters. Try a few different values of k and pick the one that seems to work best. Plot the clusters so you can see them, and write a short description of what each group looks like.

3.5 Neural Network

Build a simple neural network that does the same job as one of your earlier models. Train it, plot how the error changed during training, and compare the result with what you got before.

04 Report

Write a short report (2 to 4 pages, PDF) that covers:

- What problem you picked and why.
- What you did at each step.
- What your main results were.
- What did not work well and what you would do differently next time.

05 Presentation

Give a 10-minute talk in class about your project. Walk through your problem, what you did, and what you found. Be ready to answer a few questions.

06 Deliverables

All due Week 13.

Item	Format
Jupyter notebook (all tasks)	.ipynb
Dataset	.csv
Report	PDF, 2 to 4 pages
Slides	Any format

07 Policies

- All work must be your own.
- Cite your data source.
- Comment your code.
- You should be able to explain any part of your code when asked. Submissions that cannot be explained will not be accepted.